

Pre-registered spectral and certified mixing analysis of the male *Drosophila* central nervous system connectome

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Abstract

We report a pre-registered test of five hypotheses about the synapse-flow random walk on the connectome of the male *Drosophila* central nervous system (166,700 neurons, 124 million synapses), in which a walker moves to a postsynaptic partner in proportion to synapse number. The hypotheses concern the spectral gap and its dependence on the neck connective, the localisation of the leading modes of an input-normalised signed map, the certified mixing depth measured by the Dobrushin coefficient, its robustness to per-synapse detection confidence, and the dependence of the signed spectrum on the neurotransmitter sign convention. After public registration and a code freeze, each was tested once. Two held: the two-neuron photoreceptor modes that lead the signed spectrum are an artefact of normalising by very small inputs, removed by an input floor at the median, and they do not depend on the sign convention. Three failed: at a threshold of five synapses per connection the spectral gap is an order of magnitude smaller than predicted ($|\lambda_2| = 0.9965$) and the chain is certifiably far from mixed after 128 steps, and at no confidence level tested was the mixing depth certifiably robust. Exploratory analyses trace the slow mode to eleven neurons of the cord, seven of them mesothoracic efferent neurons, that become nearly closed once weak connections are dropped, find a similar boundary set in the lamina of the FlyWire female brain, show that null models preserving cell-type wiring reproduce the slow modes, and show that exact chains built from the confidence data change the depth profile after 32 steps by at most 0.017 at the pre-registered confidence level, far less than the worst-case certificate allows. Random-walk summaries of connectomes are dominated by nearly closed sets at the edges of the reconstruction unless these are treated explicitly.

1 Introduction

Whole-nervous-system connectomes invite dynamical summaries. Random walks on the wiring diagram identify neurons that attract or repel flow [1]; eigen-decompositions of linear models built from synapse counts propose “eigencircuits” as candidate units of computation [2]; spiking models built from synapse counts reproduce sensorimotor responses [3]; input-normalised influence models, maximum-flow analyses and layer assignments describe how sensory signals reach motor neurons [4–7]. These summaries are usually reported as point estimates computed once, without a prediction stated in advance, without a guarantee on their numerical accuracy, and without a test of how much they depend on the uncertainty of the reconstruction itself, although automated synapse detection assigns every synapse a confidence score [7, 8].

The complete connectome of the male *Drosophila* central nervous system [7, 9], which joins the brain and the ventral nerve cord (VNC), makes a whole-CNS version of these questions possible. (A female brain-and-cord connectome, BANC, appeared shortly before [6]; it was analysed with an unsigned influence model, spectral clustering and centrality, not with the mixing questions asked here.) We ask a simple question of the synapse-flow random walk, in which a walker at a

neuron moves to a postsynaptic partner with probability proportional to the number of synapses: after how many synaptic steps has the walk forgotten where it started? The quantity that answers it is the Dobrushin ergodicity coefficient $\tau(P^r)$ of the r -step chain [10, 11], the largest total-variation distance that two starting neurons can still keep after r steps [12]. The number of steps after which it becomes small is the classical counterpart of the entanglement-breaking index of a quantum channel, the number of self-compositions after which a channel forgets its input [13]. Unlike a spectral gap, $\tau(P^r)$ at finite r can be bounded from both sides by certified numerical bounds, and the bounds can be carried through a stated class of reconstruction errors.

We did this as a pre-registered study [14]. An exploratory analysis of the male CNS (19 September 2026) produced five hypotheses with numerical pass and fail criteria, about the spectral gap of the chain and its dependence on the neck connective (H1), the localisation of the leading modes of an input-normalised signed linear map (H2), the certified mixing depth at a threshold of five synapses per connection (H3), the robustness of the certified depth of the one-synapse chain to per-synapse confidence (H4), and the dependence of the signed spectrum on the neurotransmitter sign convention (H5). The pre-registration was made public on 26 September 2026 [15], the confirmatory scripts were tested on synthetic data only and frozen [16], and each hypothesis was then run once. Section 3 reports all five against their frozen criteria: H2 and H5 held, H1, H3 and H4 failed. Section 4 reports analyses that we made afterwards to understand the failures; they are exploratory and labelled as such. They show that the slowest mode of the chain at the five-synapse threshold is carried by eleven neurons of the cord, seven of them efferent neurons of the mesothoracic neuromere, that become a near-absorbing set once weak connections are dropped, that the same kind of boundary set carries the slowest mode of the FlyWire female brain at one synapse [17, 18], and that chains built from the confidence data change the depth profile far less than the worst-case certificate allows, most of whose slack lies in the step-by-step accumulation of perturbations rather than in the radius assigned to each neuron.

2 Data and methods

2.1 Connectome and neuron set

We used the male CNS connectome version 1.0, flat-connectome release at minimum synapse confidence 0.5 [7, 9] (CC-BY). The three input tables were checked against md5 sums frozen in the pre-registration. Neurons are the 166,700 bodies with a superclass annotation; the segment-to-segment table (151,856,684 rows, 311,833,243 synapses) restricted to pairs of neurons gives 25,582,938 directed edges carrying 124,177,617 synapses, 6,242,118 of them with at least five synapses. The giant strongly connected component (SCC) at weight $w \geq 1$, denoted G , has 165,314 neurons (99.2%) and 25,545,360 edges; at $w \geq 5$ the giant SCC has 157,821 neurons and 6,120,084 edges. Signs follow the consensus neurotransmitter prediction of each neuron [7]: acetylcholine +1; GABA, glutamate and histamine −1; all others 0 (base convention). The alternative convention of Pospisil et al. [2] adds dopamine +1, serotonin and octopamine −1.

2.2 Synapse-flow chain and signed map

On a giant SCC with weights w_{ij} (synapses from i to j), the synapse-flow chain is $P_{ij} = w_{ij} / \sum_k w_{ik}$, a row-stochastic matrix; a walker at i moves to a postsynaptic partner in proportion to synapse number. Its stationary distribution π was computed by power iteration and its leading eigenvalues by ARPACK [19] (implicitly restarted Arnoldi, tolerance 10^{-8} , seeded starting vectors). The signed map is $M_{ji} = s_i w_{ij} / \text{in}_j$, with s_i the sign of the presynaptic neuron and in_j the input weight of j from partners of nonzero sign (postsynaptic normalisation); the floored map of H2 replaces in_j by $\max(\text{in}_j, F)$. The participation ratio of an eigenvector v is $1 / \sum_i p_i^2$ with $p_i = |v_i|^2 / \sum_k |v_k|^2$.

2.3 Mixing depth and certified bounds

For a stochastic matrix Q , the Dobrushin coefficient is

$$\tau(Q) = \frac{1}{2} \max_{a,b} \|Q_{a\cdot} - Q_{b\cdot}\|_1 = 1 - \min_{a,b} \sum_k \min(Q_{ak}, Q_{bk}), \quad (1)$$

so that $\tau(P^r) = 0$ exactly when every starting neuron leads to the same distribution after r steps [10, 11]. Two certified bounds are cheap to compute on a chain of this size. For any set W of witness rows, $\tau(P^r) \geq \max_{a,b \in W} d_r(a, b)$ with $d_r(a, b) = \frac{1}{2} \|(e_a - e_b)P^r\|_1$; and for any set K of columns, $\tau(P^r) \leq 1 - \sum_{k \in K} \min_i (P^r)_{ik}$ [11]. Witnesses were 90 random rows and the 10 rows of largest stationary mass (seed 7); hub columns were the 500 columns of largest stationary mass. Rounding was bounded a priori [20]: after r steps each propagated row carries an ℓ_1 error of at most about $r(D+1)u$, with D the largest in- or out-degree and $u = 2^{-53}$, and the final sums add at most about $2nu$; a slack of $10^{-9}r$, subtracted from lower and added to upper bounds, covers both (and the rounding of the robust sums below) with a margin of at least a factor of ten at every r . The bounds are therefore statements about the exact chain defined by the data; in the text and tables they are rounded to the digits shown, and the result files give them in full.

2.4 Perturbation classes

A perturbation class is a set of chains P' that the data cannot rule out. The *uniform* class $C(\delta)$ multiplies every edge weight by a factor in $[1 - \delta, 1 + \delta]$. The *confidence* class at level c allows every edge weight w'_{ij} between $w_{ij}^{\text{hi}}(c)$, the number of its synapses with $\min(\text{conf}_{\text{pre}}, \text{conf}_{\text{post}}) \geq c$, and w_{ij} ; that is, any subset of the synapses below confidence c may be false detections. Both classes are handled by the following two facts (proofs in Appendix A).

Lemma 1. *If $0 \leq w' \leq w$ entrywise on a row with total weights $W' = \sum_j w'_j > 0$ and $W = \sum_j w_j$, the normalised rows satisfy $\text{TV}(w/W, w'/W') \leq (W - W')/W$.*

Lemma 2. *Let P, P' be stochastic with $\text{TV}(P_i, P'_i) \leq \varepsilon_i$ for every row i , and $x_t = (e_a - e_b)P^t$. Then $|d_r(a, b; P') - d_r(a, b; P)| \leq \sum_{t=0}^{r-1} \sum_i |x_t(i)| \varepsilon_i$.*

For the confidence class, Lemma 1 gives $\varepsilon_i(c) = 1 - W_i^{\text{hi}}(c)/W_i$, the fraction of the row's synapses below confidence c , and Lemma 2 turns every witness pair into a lower bound on $\tau(P'^r)$ valid for the whole class. The exploratory analysis had used this argument without the $t = 0$ term; that error, found before registration, is corrected throughout (Section 2.5).

2.5 Pre-registration and protocol

The pre-registration was drafted and frozen on 19 September 2026 after an exploratory phase in which every number of its Section 4 was computed; it was first made public, byte-identical, on 26 September 2026 [15], which is its registration date. Before any confirmatory computation, a dated log of deviations and clarifications fixed every definition the text left open (the removal sets of H1, the floor of H2, the graph, witnesses and confidence grid of H4, the carrier sets of H5, the eigen-solver seeds and the run protocol), and the confirmatory scripts were tested only on synthetic data against independent dense computations and frozen [16]. The pre-registration text contained three errors, found and logged before registration: the robust lower bounds printed in its Section 4.4 had been computed without the $t = 0$ term of Lemma 2 and were too high by 0.21; its knock-out of ascending and descending neurons removed 3,693 neurons, not the 3,160 stated; and its claim that no brain-and-cord connectome had been analysed before overlooked BANC [6]. The analysis ran once, on 26 September 2026, on a Linux container (two cores, 7.8 GB) with Python 3.11, NumPy 2.4.4, SciPy 1.17.1 and one BLAS thread, after a check that every executed file matched the frozen manifest; every script ran once, without error.

Table 1: The five pre-registered hypotheses, each run once after the freeze. $|\lambda_2|$: modulus of the second eigenvalue of the synapse-flow chain on the giant SCC; ρ : spectral radius of the signed map; PR: participation ratio; $\tau(P^r)$: Dobrushin coefficient of the r -step chain.

	Criterion (frozen)	Value	Met
H1 (i)	$ \lambda_2 $ at $w \geq 5$ lies in $[0.90, 0.97]$	0.996524	no
H1 (ii)	removing ascending, descending and sensory-ascending neurons gives $1 - \lambda_2 < 0.002$	0.000190	yes
H1 (iii)	removing the motor neurons moves $ \lambda_2 $ by less than 0.01	0.000726	yes
H1			fail
H2 (a)	with the floor F (median input, 342 synapses), $\rho < 0.85$	0.83484	yes
H2 (b)	at least 10 of the top 20 modes have PR > 20	19 of 20	yes
H2 (c)	descriptive: a top-20 mode with more than half its power on lobula-plate or ellipsoid-body neurons	mode 1, 99.3% on the ellipsoid body	reported
H2			pass
H3 (1)	certified lower bound on $\tau(P^{32})$ at $w \geq 5$ exceeds 0.20	0.9229	yes
H3 (2)	first checkpoint with lower bound < 0.05 lies in $[48, 96]$	none up to 128 (0.6695 at 128)	no
H3			fail
H4	certified lower bound on $\tau(P^{32})$ over the confidence class at $c = 0.70$ exceeds 0.10	no certificate (-0.52)	no
H4 (clause)	largest level $c \in [0.51, 0.95]$ at which the bound at $r = 32$ exceeds 0.10	none (best: 0.071 at $c = 0.51$)	reported
H4			fail
H5	ρ changes by less than 0.02 under the Pospisil convention	0.000015	yes
H5	the carriers of the top six modes are unchanged	same six neurons	yes
H5	sensitivity result		pass

3 Confirmatory results

Table 1 lists every pre-registered criterion with the value obtained and the verdict. The reference values of the exploratory phase (at $w \geq 1$) were reproduced to every digit printed in the pre-registration before the confirmatory run.

H1: the gap at five synapses is an order of magnitude smaller than predicted. At $w \geq 1$ the chain has $|\lambda_2| = 0.94503$, a spectral gap of 0.055. At $w \geq 5$ the prediction was a gap between 0.03 and 0.10; the observed value is $|\lambda_2| = 0.996524$, a gap of 0.0035. The two knock-out predictions held: without the ascending, descending and sensory-ascending neurons the giant SCC (153,784 neurons) is still two-sided, held together by 18 residual neck neurons, and has $1 - |\lambda_2| = 0.00019$; without the motor neurons $|\lambda_2|$ moves by 0.0007.

H2 and H5: the photoreceptor modes are a normalisation effect. The leading eigenvalues of the signed map come in near-equal pairs of opposite sign ($-0.9151, +0.9137, -0.9122, +0.9098$), each mode carried by a pair of R7/R8 photoreceptors (participation ratio 2.0 to 2.4). These photoreceptors receive few synapses (the six carriers have 2 to 43 input synapses from partners of nonzero sign, against 63 to 174 outputs), so postsynaptic normalisation turns a reciprocal pair into a two-neuron loop of gain close to one. With the input floored at the median,

Table 2: H4: certified lower bounds on $\tau(P'^r)$ over the confidence class at level c , and over the uniform class $C(0.05)$, with 100 witness rows on G . A negative value means that no certificate is obtained. The last row is the nominal bound for P itself.

r	1	2	4	8	16	32	48
$c = 0.51$	1.000	0.997	0.977	0.860	0.555	0.071	-0.015
$c = 0.55$	1.000	0.984	0.894	0.595	-0.028	-0.071	-0.071
$c = 0.60$	1.000	0.941	0.759	0.264	-0.192	-0.198	-0.199
$c = 0.70$	1.000	0.821	0.396	-0.470	-0.511	-0.520	-0.522
$C(0.05)$	0.789	0.579	0.156	-0.441	-0.481	-0.502	-0.508
nominal P	1.000	1.000	0.998	0.936	0.707	0.312	0.127

$F = 342$, the spectral radius falls to 0.83484 and 19 of the 20 leading modes extend over more than 20 neurons. The leading floored mode is carried by GABAergic ellipsoid-body ring neurons (ER3d_b and ER3p_a, participation ratio 96.6, 99.3 % of its power on neurons whose primary neuropil is the ellipsoid body), and it is the ninth mode of the unfloored map with the same eigenvalue: the floor removes the photoreceptor loops above it and leaves it untouched. Six of the next eight modes are ring-neuron modes as well (ER4d, ER3p_a, ER3m and three on ER3w_b) and two are on the leg sensory neurons LgAG1; the remaining modes of the top 20 are on Kenyon cells, antennal-lobe projection neurons and sensory neurons of the cord. The result is the same with the floor at 700 synapses (18 of 20 modes extended). Under the Pospisil sign convention the unfloored spectral radius changes by 1.5×10^{-5} and the same six neurons carry the top six modes (H5).

H3: certified slow mixing. At $w \geq 5$ the certified bounds are $0.923 \leq \tau(P^{32}) \leq 0.967$ and $0.669 \leq \tau(P^{128}) \leq 0.816$ (Figure 1): after 128 synaptic steps two starting neurons can still be told apart with total-variation distance at least 0.669. The first criterion held and the second failed, because the bound never dropped below 0.05. This is consistent with H1: at $|\lambda_2| = 0.9965$ a deviation decays by a factor ten only after several hundred steps.

H4: the worst-case certificate does not survive the confidence class. All 123,954,262 synapse-partner rows inside G matched the connection weights exactly, and none had a confidence below 0.5. On average 14.4 % of a neuron’s output synapses (12.3 % weighted by the stationary distribution) have $\min(\text{conf}_{\text{pre}}, \text{conf}_{\text{post}}) < 0.7$. With per-row radii of this size, the certified lower bound of Lemma 2 is 0.821 at $r = 2$ and 0.396 at $r = 4$ and gives no certificate from $r = 8$ on (Table 2). Even at $c = 0.51$, where on average 0.5 % of a neuron’s synapses lie below the level, the bound is 0.071 at $r = 32$, short of the pre-registered 0.10. As the pre-registration requires us to state: the depth result is not certifiably robust to reconstruction uncertainty at the 0.7 level, nor at any level tested.

4 Exploratory analyses

Everything in this section was computed after the confirmatory run, to understand its outcome. None of it was pre-registered, and none of it changes a verdict of Table 1. The scripts, logs and result files are in the repository under `code/exploratory` and `exploratory/`.

4.1 What carries the slow mode

For the chain on each giant SCC we computed the left and right eigenvectors of λ_2 and the sweep cut of smallest conductance along the right eigenvector, a standard way of locating a metastable

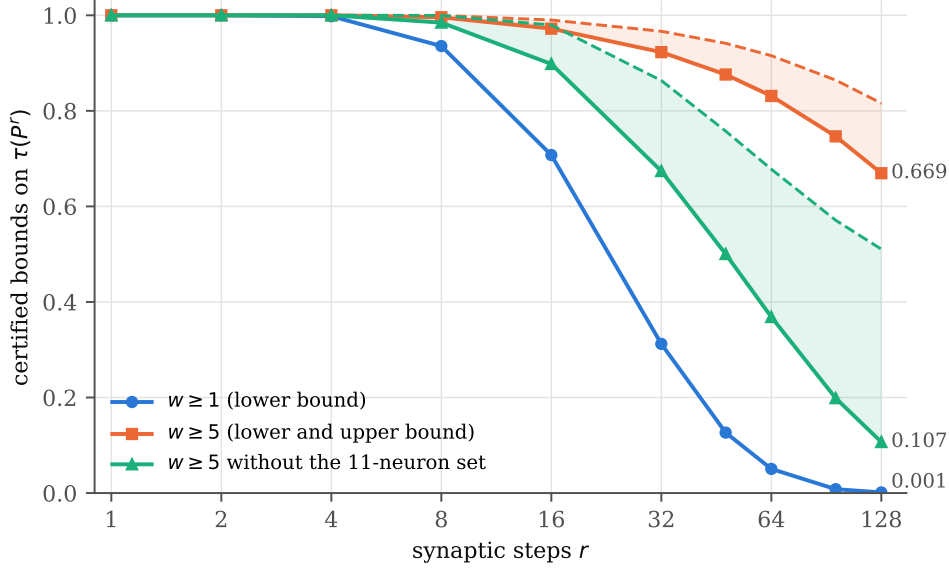


Figure 1: Certified bounds on the Dobrushin coefficient $\tau(P^r)$ of the synapse-flow chain of the male CNS: the largest total-variation distance between two starting neurons after r synaptic steps. Blue: witness lower bound at $w \geq 1$ (the nominal chain of H4); orange: lower (solid) and upper (dashed) bounds at $w \geq 5$ (H3); green: the same at $w \geq 5$ with the eleven-neuron set of Table 3 removed (exploratory, Section 4.1). Every bound holds for the exact chain, with rounding accounted for; the true value lies between the lower and upper curve.

set [21, 22]: the set S of neurons with the smallest (or largest) eigenvector entries that minimises $\phi(S) = F(S \rightarrow S^c) / \min\{\pi(S), \pi(S^c)\}$, with F the stationary probability flow per step.

At $w \geq 5$ the slow mode is carried by eleven neurons of the VNC (Table 3; VNC type names follow the systematic nomenclature of the male adult nerve cord connectome [23–25]): seven efferent neurons with somata on the midline of the mesothoracic neuromere (EN00B001, EN00B008, two EN00B011, two EN00B015 and the ventral unpaired median neuron mesVUM-MJ), two IN03B088 and one IN19A061 interneuron, and one abdominal motor neuron, MNad21. Together they hold 12.8% of the stationary mass of the chain, a walker leaves them with probability 3.5×10^{-3} per step, and the two-block estimate $1 - p - q = 0.99598$ reproduces $|\lambda_2| = 0.99652$. The left eigenvector puts 99.7% of its negative mass in the cord. The eleven neurons are not a trap at $w \geq 1$: there they hold 0.4% of the stationary mass and a walker leaves them with probability 0.29 per step. What changes is their output. At $w \geq 1$ they send 299 synapses to the other neurons of G , only 43 of them on connections of five or more synapses (six connections), while they receive 21,846 synapses, 19,378 of them on such connections, and 514 synapses connect them with each other. Dropping weak connections removes most of the way out and little of the way in. Efferent neurons send their main output out of the CNS, to targets that are not in the connectome, so the within-CNS normalisation of the chain leaves them with few, mostly local, exits. This set also sets the certified bound of H3: from $r = 4$ on, the witness pair that attains the bound has one member inside it.

Removing the eleven neurons lowers $|\lambda_2|$ at $w \geq 5$ to 0.9806, and the next metastable set is again small and in the cord (five VNC interneurons and one motor neuron, 4.0% of the stationary mass, conductance 0.027). The certified bounds become $0.674 \leq \tau(P^{32}) \leq 0.863$ and $0.107 \leq \tau(P^{128}) \leq 0.511$. Removing every efferent superclass instead (24 neurons of the SCC) gives nearly the same numbers ($|\lambda_2| = 0.9806$, the same six-neuron set next). At the five-synapse threshold, then, the slowest dynamics of the synapse-flow chain are set by a succession of small, nearly closed sets in the cord, not by the neck connective (Figure 1).

Table 3: The slow mode of the synapse-flow chain. Cut: smallest-conductance sweep set along the slow right eigenvector; mass: its stationary probability; escape: probability per step that a walker in the set leaves it (stationary flow divided by mass).

Chain	$ \lambda_2 $	cut size	mass	escape	carried by
male CNS, $w \geq 1$	0.94503	727	0.0047	0.116	central complex, anterior visual pathway
male CNS, $w \geq 5$	0.99652	11	0.1276	0.0035	mesothoracic efferent neurons
male CNS, $w \geq 5$, without the 11	0.98062	6	0.0398	0.027	VNC interneurons, one motor neuron
FlyWire brain, $w \geq 1$	0.99999	7	0.0001	2.8×10^{-5}	photoreceptors R1–6, L2
FlyWire brain, $w \geq 5$	0.95463	891	0.0177	0.069	anterior visual pathway, central complex
male brain only, $w \geq 1$	0.92559	1,416	0.0179	0.114	anterior visual pathway, central complex
male brain only, $w \geq 5$	0.93179	1,495	0.0150	0.105	anterior visual pathway, central complex

At $w \geq 1$ the picture of the exploratory phase holds for the left eigenvector (94.3 % of one sign in the brain, 92.9 % of the other in the cord), but no small set dominates: the sweep cut of smallest conductance is a set of 727 neurons of the central complex and the anterior visual pathway (414 with primary neuropil in the ellipsoid body; medulla-tubercle, tubercle-bulb, ring, EPG, PEN and PEG neurons [26]), with conductance 0.116 and a two-block estimate of 0.883 that does not reproduce $|\lambda_2| = 0.945$.

4.2 Is the failure of H4 the certificate or the data?

Lemma 2 bounds every chain of the confidence class at once, by a first-order argument that lets each step take its own worst case. To see how far the depth numbers actually move, we computed specific chains exactly. $M(c)$ removes every synapse with confidence below c , a vertex of the class at level c (a neuron whose outputs would all be removed keeps them, as the class allows); $B(k)$ keeps each synapse with probability equal to the midpoint of its confidence bin, a random reconstruction that is not a member of the class. For each chain we recomputed the distances d_r of all 4,950 pairs of the H4 witness rows and $|\lambda_2|$ at both thresholds (Table 4).

The chains move little (Table 4, Figure 2). Removing every synapse below confidence 0.70 changes a neuron’s output distribution by 0.080 in total variation on average; it changes no witness distance at $r = 32$ by more than 0.017 (and none at any $r \leq 64$ by more than 0.043), and $\max d_{32}$ falls from 0.312 to 0.296. Removing every synapse below 0.90, half of all synapses, lowers $\max d_{32}$ to 0.264. The five random reconstructions remove 15 % of the synapses and change no witness distance at $r = 32$ by more than 0.016 (0.11 at $r = 8$), while $\max d_r$ moves by less than 0.01 at every r . $M(0.60)$ and $M(0.70)$ are members of the class at $c = 0.70$, for which the certified bound of H4 is negative from $r = 8$ on. The second eigenvalue at $w \geq 1$ stays within 0.008 of its value in every chain. At $w \geq 5$ it follows the eleven-neuron set of Section 4.1. In $M(0.80)$ and $M(0.90)$ the set keeps 15 and none of the 43 synapses on its connections of five or more synapses to the other neurons of G , leaves the giant SCC (one of its neurons remains in $M(0.80)$), and $|\lambda_2|$ falls to 0.940 and 0.952. The random reconstructions keep 21 to 35 of those synapses; where the set stays in the giant SCC with most of its stationary mass ($B(2)$ and $B(4)$, 8 % and 11 %) $|\lambda_2|$ is 0.994 and 0.996, and where it is broken up (two to six of its neurons left, with mass at most 0.0014) $|\lambda_2|$ is 0.967 to 0.976, still above its value at $w \geq 1$. The slowest mode at five synapses is thus a property of a few connections that detection errors can make or break.

Where the certificate loses can be located. Lemma 1 charges row i the fraction ε_i of its

Table 4: Chains built from the confidence data (exploratory). $M(c)$: every synapse with confidence below c removed; $B(k)$: every synapse kept with probability equal to the midpoint of its confidence bin (five draws). Synapses: total weight kept; row TV: mean total-variation change of a neuron’s output distribution; set: how many of the eleven neurons of Table 3 are in the giant SCC at $w \geq 5$; $\max d_{32}$: the witness lower bound on $\tau(P'^{32})$ for that chain; Δ_{32} : largest change of d_{32} over the 4,950 witness pairs, relative to P .

Chain	synapses (10^6)	row TV	$ \lambda_2 , w \geq 1$	$ \lambda_2 , w \geq 5$	set	$\max d_{32}$	Δ_{32}
P	123.95	0.000	0.9450	0.9965	11	0.312	
$M(0.60)$	116.58	0.042	0.9441	0.9964	11	0.304	0.008
$M(0.70)$	106.16	0.080	0.9432	0.9963	11	0.296	0.016
$M(0.80)$	86.76	0.134	0.9413	0.9403	1	0.286	0.048
$M(0.90)$	60.84	0.210	0.9374	0.9521	0	0.264	0.074
$B(1)$	105.37	0.078	0.9440	0.9763	2	0.304	0.015
$B(2)$	105.36	0.078	0.9439	0.9944	8	0.303	0.014
$B(3)$	105.36	0.078	0.9439	0.9738	3	0.305	0.015
$B(4)$	105.36	0.078	0.9438	0.9962	11	0.303	0.012
$B(5)$	105.36	0.078	0.9441	0.9669	6	0.306	0.010

synapses below the level, which is the largest possible change of the row only if all of them go to partners that receive no high-confidence synapse from i . The exact largest change over the class has a closed form (proof in Appendix A).

Lemma 3. *For a row with partner weights w_j , high-confidence parts w_j^{hi} , $l_j = w_j - w_j^{\text{hi}}$ and $W = \sum_j w_j$, the largest total-variation change of the normalised row over the confidence class is*

$$\varepsilon^* = \max_S \left[\frac{w(S)}{W} - \frac{w^{\text{hi}}(S)}{W - l(S)} \right] \leq \varepsilon,$$

the maximum over nonempty sets S of partners (sums over S), attained by removing every low-confidence synapse to S and none to the other partners. (A row without high-confidence synapses may be emptied, and then $\varepsilon^ = 1$.)*

We computed ε^* exactly for rows with at most 12 partners and bounded it from above, still rigorously, for the others. At $c = 0.70$ the mean radius falls from 0.144 to 0.089, close to the 0.080 by which the vertex $M(0.70)$ actually moves a row, but the certificate gains little depth (Table 5): at $r = 4$ the bound rises from 0.396 to 0.621, at $r = 8$ it remains negative. At $c = 0.55$ it now certifies $\tau(P'^{16}) \geq 0.106$ for the whole class, where the radius of H4 gave nothing. At $r = 32$ the best certificate, at $c = 0.51$, rises from 0.071 to 0.096, still short of the pre-registered 0.10, and from $c = 0.52$ on there is none. Most of the slack is in Lemma 2. Evaluated with the actual row changes of $M(0.70)$, which is not a certificate but the most favourable radius for that one chain, the lemma gives 0.026 at $r = 8$, where $M(0.70)$ itself has 0.934. The lemma adds, at every step, the perturbations of all the rows that the two walkers visit, as if every one of them pushed the walkers apart; in the actual chain they point in different directions and largely cancel.

4.3 Null models

Each null model keeps every connection’s presynaptic neuron and weight and re-draws its postsynaptic partner by permuting targets within a group of connections, so out-degrees, out-strengths and the multiset of in-connections are kept. N1 permutes over all connections; N2 within each pair (superclass of source, superclass of target); N3 within each pair of cell types (three draws each, Table 6). At $w \geq 1$ the configuration null N1 mixes fast ($|\lambda_2| \approx 0.15$, 9 % of

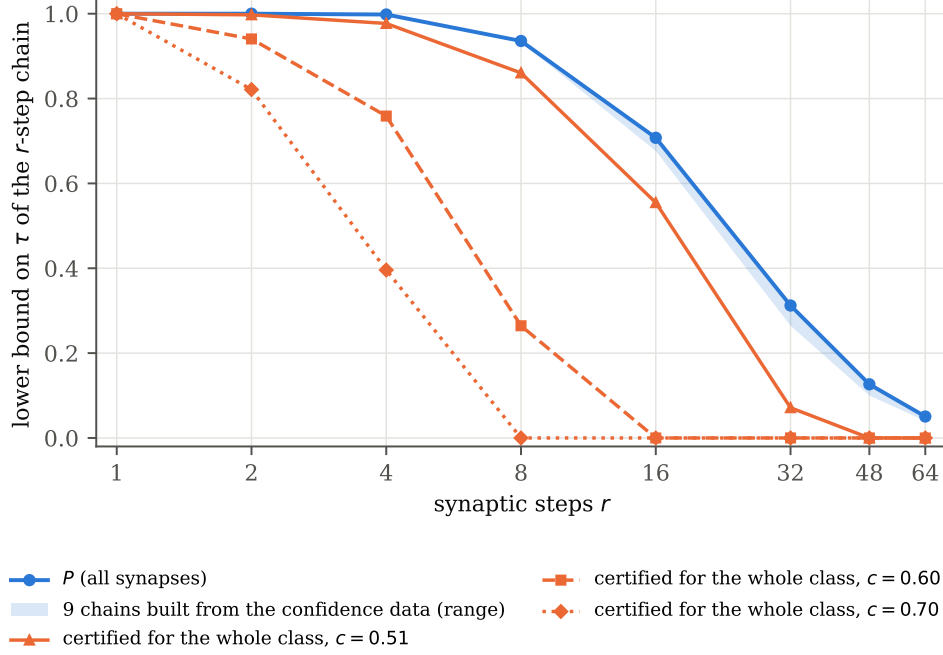


Figure 2: The certificate of H4 against exact chains (exploratory). Blue line: witness lower bound $\max_{a,b} d_r(a,b)$ on $\tau(P^r)$ for the chain P of all synapses ($w \geq 1$, the 4,950 pairs of the H4 witness rows); blue band: its range over the nine chains $M(0.60)$ to $M(0.90)$ and $B(1)$ to $B(5)$ of Table 4. Orange: the certified lower bound of H4 (Lemma 2), valid for every chain of the confidence class at level c , drawn at zero where it is negative, that is, where no certificate is obtained. The chains $M(0.60)$ and $M(0.70)$ belong to the class at $c = 0.70$.

the stationary mass on the top 1 % of neurons against 42 % observed); preserving superclass-to-superclass counts restores most of the slow mode ($|\lambda_2| = 0.928$), and preserving type-to-type counts restores all of it (0.9446 to 0.9447 against 0.9450, with a smallest-conductance cut of 714 to 716 neurons against 727 observed, and 41 % of the stationary mass on the top 1 %). At $w \geq 5$ the superclass null gives 0.91 to 0.94; the type null reproduces a trap of 11 to 13 neurons in two of three draws ($|\lambda_2| = 0.998$ and 0.996) and not in the third (0.953). The slow modes are thus a property of cell-type wiring, and at $w \geq 5$ they depend on the particular connections of a few efferent types.

4.4 The FlyWire female brain

The same chain on the FlyWire whole-brain connectome of a female fly [17, 18] (version 783, 139,255 proofread neurons, 54.5 million synapses [27]) and on the male CNS with the cord removed (the brain-only row of H1) gives the numbers of Tables 3 and 7. Two things agree between the datasets and one does not.

First, the slowest well-populated subsystem is the same in both brains: neurons of the anterior visual pathway and the central complex [26]. In FlyWire at $w \geq 5$ the cut (891 neurons) is led by medulla-tubercle (MeTu), tubercle-bulb (TuBu), ring (ER) and EPG neurons and Dm-DRA1; in the male brain (1,416 and 1,495 neurons) by MeTu neurons, dorsal-rim R7 photoreceptors, Dm-DRA1 and EPG neurons, with Delta7 neurons at one synapse and Mi15 medulla neurons at five. The conductance of these sets is 0.07 to 0.11. The pathway from the dorsal rim to the ellipsoid body carries sky-compass information; in the synapse-flow chain it is a weakly coupled channel. Second, the certified depth profiles are of the same order: the lower bound on $\tau(P^{32})$ is 0.28 (FlyWire, $w \geq 1$), 0.26 (FlyWire, $w \geq 5$), 0.11 and 0.17 (male brain), against 0.31 for the

Table 5: Where the certificate of H4 loses (exploratory). Certified lower bounds on $\tau(P^r)$ for every chain of the confidence class at level c , with the row radius of H4 and with the radius of Lemma 3 (exact for rows with at most 12 partners, a certified upper bound otherwise); for comparison, Lemma 2 evaluated with the actual row changes of the vertex $M(c)$ (valid for that chain only), and the exact witness bound of $M(c)$ from Table 4. Mean: mean radius over the rows of G . A negative value means that no bound is obtained.

	radius	mean	$r = 2$	$r = 4$	$r = 8$	$r = 16$	$r = 32$
$c = 0.55$	ε (Lemma 1, H4)	0.028	0.984	0.894	0.595	-0.028	-0.071
	ε^* (Lemma 3)	0.023	0.987	0.916	0.665	0.106	-0.056
	row change of $M(c)$, not certified	0.023	0.987	0.918	0.671	0.119	-0.055
$c = 0.60$	ε (Lemma 1, H4)	0.059	0.941	0.759	0.264	-0.192	-0.198
	ε^* (Lemma 3)	0.043	0.966	0.832	0.448	-0.140	-0.152
	row change of $M(c)$, not certified	0.042	0.970	0.840	0.469	-0.132	-0.143
	exact witness bound of $M(c)$		1.000	0.998	0.935	0.702	0.304
$c = 0.70$	ε (Lemma 1, H4)	0.144	0.821	0.396	-0.470	-0.511	-0.520
	ε^* (Lemma 3)	0.089	0.913	0.621	-0.065	-0.304	-0.321
	row change of $M(c)$, not certified	0.080	0.930	0.657	0.026	-0.265	-0.281
	exact witness bound of $M(c)$		1.000	0.998	0.934	0.697	0.296
nominal P			1.000	0.998	0.936	0.707	0.312

Table 6: Null models: $|\lambda_2|$ of the giant SCC (three draws each) and stationary mass of the top 1 % of neurons.

	observed	N1 (configuration)	N2 (superclass blocks)	N3 (type blocks)
$w \geq 1, \lambda_2 $	0.945	0.155, 0.152, 0.152	0.928, 0.928, 0.928	0.945, 0.945, 0.945
$w \geq 1$, top 1 % mass	0.42	0.09	0.19	0.41
$w \geq 5, \lambda_2 $	0.997	0.230, 0.224, 0.225	0.935, 0.913, 0.913	0.998, 0.996, 0.953
$w \geq 5$, top 1 % mass	0.73	0.13	0.27	0.72, 0.69, 0.63

whole male CNS at $w \geq 1$.

What does not agree is $|\lambda_2|$. In FlyWire at $w \geq 1$ it is 0.99999, set by seven neurons (four R1-6 photoreceptors, one L2 and two unlabelled optic-lobe neurons) that hold about 10^{-4} of the stationary mass and are left with probability 2.8×10^{-5} per step. Like the efferent neurons of the male cord, the set sits at an edge of the reconstruction, here the lamina. It dominates $|\lambda_2|$ but, carrying almost no stationary mass, has no visible effect on the certified depth profile. The second eigenvalue of a connectome random walk is a statement about its most nearly closed set, however small; the depth profile is a statement about where the walk actually goes.

5 Discussion

What the pre-registered tests established. Two predictions held. The two-neuron photoreceptor modes that lead the spectrum of the input-normalised signed map are an artefact of dividing by very small inputs (H2), and they do not depend on the sign convention (H5). A floor at the median input removes them and leaves extended modes of GABAergic ellipsoid-body ring neurons in the lead, the kind of circuit that Pospisil et al. found among the eigencircuits of the unnormalised female-brain matrix [2]. Linear models and graph traversals that normalise by postsynaptic input [4, 6] inherit this sensitivity at their low-input neurons, which are often sensory; reporting the input weights of the carriers of any leading mode is a cheap guard. Three predictions failed. The spectral-gap and depth predictions at the five-synapse threshold (H1 (i), H3) were set by values of the weak-threshold chain, and the chain at five synapses turned out to

Table 7: Brain-only chains: certified lower bounds on $\tau(P^r)$ (100 witness rows, seed 7) and the stationary mass of the top 1% of neurons. The whole male CNS at $w \geq 1$ is shown for comparison.

Chain	neurons	$r = 8$	$r = 16$	$r = 32$	$r = 64$	top 1% mass
FlyWire brain, $w \geq 1$	135,403	0.897	0.640	0.285	0.043	0.41
FlyWire brain, $w \geq 5$	119,756	0.933	0.631	0.261	0.050	0.56
male brain only, $w \geq 1$	145,035	0.854	0.451	0.112	0.007	0.35
male brain only, $w \geq 5$	139,150	0.895	0.549	0.165	0.016	0.48
male CNS, $w \geq 1$	165,314	0.936	0.707	0.312	0.051	0.42

be an order of magnitude slower; the two knock-out predictions of H1 (ii) and (iii) held. The certificate of robustness to reconstruction uncertainty (H4) failed at every confidence level tested.

Boundary sets dominate slow modes. The exploratory analyses give one explanation for both depth failures, and it is a caution for any random-walk analysis of a connectome. A random walk normalised within the reconstructed graph treats every neuron as if all its outputs were in the graph. Efferent and motor neurons, whose main targets are outside the CNS, and neurons at an incompletely reconstructed edge of the volume, are left with a few within-graph exits; once weak connections are dropped these can close almost completely. Eleven neurons of the male cord, seven of them mesothoracic efferent neurons, do this at five synapses, and seven neurons at the lamina of the FlyWire brain do it at one. Such sets set $|\lambda_2|$ and any mixing time derived from it, whatever their size, and they set certified depth bounds when they carry stationary mass. Three remedies follow: add an explicit exit state that receives the synapses a neuron makes outside the graph (the male CNS tables record synapses to non-neuron bodies, which makes this possible), report results with boundary classes removed, and report depth profiles with the witness sets and stationary masses that attain them rather than a single eigenvalue. The threshold matters as much: the same connectome gives $|\lambda_2| = 0.945$ at one synapse and 0.9965 at five.

Certificates and reconstruction uncertainty. The failure of H4 is not evidence that the depth profile is fragile. Every chain that we built from the confidence data, including vertices of the class that remove every synapse below the level, moved the witness distances after 32 steps by at most 0.017 at the pre-registered level, and by less than 0.08 even after half of all synapses were removed, where the certificate allows changes of order one. Sharpening the row radius (Lemma 3) recovers only part of the gap; most of it is lost in the step-by-step accumulation of Lemma 2, which cannot use the cancellation between the perturbations of different rows. The certificate cannot rule out that some adversarial member of the class moves the profile much further, and whether one exists is open. A useful certificate at the depths that matter would have to exploit that cancellation, or restrict the class to what detection errors can plausibly do; until then, robustness of connectome summaries to reconstruction uncertainty can be shown by resampling, as in Table 4, but not certified.

Limitations. The confidence class models false detections only; missing synapses, which incomplete proofreading makes common [7], are not in it, and a class that allowed them would be larger. Only the bounds on $\tau(P^r)$ are certified; eigenvalues were computed by ARPACK to a tolerance of 10^{-8} and are not. The chains are not reversible, so the sweep cuts locate metastable sets heuristically [21, 22]. The FlyWire comparison differs in sex, reconstruction pipeline and synapse detection. Most of all, the synapse-flow chain is a structural object: it says how a walker that follows synapses spreads over the wiring diagram, not how activity spreads in a fly.

Data and code availability

The male CNS connectome v1.0 is public under CC-BY [7, 9]; the md5 sums of the four input tables used here are recorded in the repository. The FlyWire connectivity data are on Zenodo [27] and the FlyWire annotations in the flywire_annotations repository [18]. The pre-registration, the deviation log, the exploratory and confirmatory code, the tests, the logs of the confirmatory run, all result files and the exploratory analyses of Section 4 are at <https://github.com/erankopel/maleCNS-depth>; the registration record and the frozen confirmatory analysis are archived at doi:10.5281/zenodo.22980248 [15] and doi:10.5281/zenodo.22981189 [16].

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Use of generative AI. This study and manuscript were prepared with the help of Claude (Anthropic). It was used to write and run the exploratory, confirmatory and test scripts, to review the confirmatory scripts before the freeze, to prepare the pre-registration record, the deviation log and the reproduction and results notes, to check the references against Crossref, DataCite and the publishers’ pages, and to draft and revise the manuscript text. The author designed the study, approved every pre-registration decision, directed and checked the work, verified the results, and takes full responsibility for the content.

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A Proofs

Lemma 1. Write $p = w/W$ and $p' = w'/W'$. Since $W' \leq W$, $p'_j = w'_j/W' \geq w'_j/W$, hence $p_j - p'_j \leq (w_j - w'_j)/W$ for every j . The total-variation distance is the sum of the positive parts of $p_j - p'_j$, so $\text{TV}(p, p') \leq \sum_j (w_j - w'_j)/W = (W - W')/W$. If $W' = 0$ the row of P' may be any distribution and $\text{TV} \leq 1 = (W - W')/W$ holds trivially. With $W' \geq W^{\text{hi}}(c)$ for every member of the confidence class, $\varepsilon_i(c) = 1 - W_i^{\text{hi}}(c)/W_i$ bounds the row change of every member. For the uniform class $C(\delta)$ every ratio P'_{ij}/P_{ij} lies in $[(1 - \delta)/(1 + \delta), (1 + \delta)/(1 - \delta)]$, which gives $\text{TV} \leq 2\delta/(1 + \delta)$; the pre-registered, slightly larger constant $\varepsilon = 2\delta/(1 - \delta)$ was kept.

Lemma 2. Let $x'_t = (e_a - e_b)P'^t$ and $\Delta_t = x'_t - x_t$, so $\Delta_0 = 0$ and $\Delta_{t+1} = \Delta_t P' + x_t(P' - P)$. Because P' is stochastic, $\|yP'\|_1 \leq \|y\|_1$ for every row vector y , and therefore $\|\Delta_r\|_1 \leq \sum_{t=0}^{r-1} \|x_t(P' - P)\|_1$. Each term satisfies $\|x_t(P' - P)\|_1 \leq \sum_i |x_t(i)| \|P'_i - P_i\|_1 \leq \sum_i |x_t(i)| 2\varepsilon_i$. Since $d_r = \frac{1}{2}\|x_r\|_1$, the triangle inequality gives $|d_r(P') - d_r(P)| \leq \frac{1}{2}\|\Delta_r\|_1 \leq \sum_{t=0}^{r-1} \sum_i |x_t(i)| \varepsilon_i$. The $t = 0$ term, $\varepsilon_a + \varepsilon_b$, is the one the exploratory script had omitted; the test suite of the repository shows that without it the bound is violated by chains sampled from both classes.

Lemma 3. For every member w' of the class, $\text{TV}(p, p') = \max_S (p(S) - p'(S))$ over sets S of partners. For a fixed S , $p'(S) = w'(S)/(w'(S) + w'(S^c))$ increases with $w'(S)$ and decreases with $w'(S^c)$, so its smallest value over the class is $w^{\text{hi}}(S)/(w^{\text{hi}}(S) + w(S^c)) = w^{\text{hi}}(S)/(W - l(S))$, attained by removing every low-confidence synapse to S and none to S^c . Taking the maximum over S gives ε^* ; and since $W - l(S) \leq W$, $w(S)/W - w^{\text{hi}}(S)/(W - l(S)) \leq l(S)/W \leq \varepsilon$. The maximum over the $2^d - 1$ nonempty sets was computed exactly for rows with $d \leq 12$ partners (3.6% of the rows of G). For the others we used an upper bound: for a fixed removed weight $C = l(S)$ the maximisation is a knapsack problem in which a partner j contributes $w_j/W - w_j^{\text{hi}}/(W - C)$ at cost l_j ; the ratio of the two decreases with the high-confidence fraction $q_j = w_j^{\text{hi}}/w_j$ for every C , so the fractional relaxation is solved, for all C at once, by taking partners in increasing order of q_j , and its largest value along this path bounds ε^* from above. On random test rows the bound exceeded the exact value by at most 0.0053 for 13 to 15 partners.